ERA: A Three-Level Framework for Evaluating Representational Alignment in Fine-Tuned Language Models

Alexander Paolo Zeisberg Militerni
alexander.zeisberg85@gmail.com
ERA Framework v1.0.1 https://github.com/blacklotus1985/ERA-framework

Abstract

Fine-tuning a language model can substantially alter its output distributions while leaving its internal concept representations largely unchanged. We term this condition *shallow alignment* and argue that standard output-level evaluation methods cannot detect it, because they do not inspect the internal model state. We present ERA (Evaluation of Representational Alignment), a diagnostic framework that decomposes fine-tuning effects into three distinct levels: behavioural drift (L_1), probabilistic drift (L_2), and representational drift (L_3). We further introduce the Alignment Score AS $= L_2/L_3$ as a scalar indicator of learning depth, and the Stereotype Index (SI) for quantifying differential gender associations across semantic role families. ERA is released as an open-source Python package and is intended to complement existing benchmarks with model-internal diagnostics unavailable from output-level analysis alone.

1 Introduction

The deployment of fine-tuned language models in consequential settings has renewed interest in methods for detecting and characterising bias. The dominant evaluation paradigm operates at the output level: a model is prompted and its responses are observed. WEAT [Caliskan et al., 2017] measures cosine similarity between concept and attribute word embeddings in a static embedding space. StereoSet [Nadeem et al., 2021] evaluates model preference for stereotypical versus anti-stereotypical sentence completions. WinoBias [Zhao et al., 2018] and WinoGender [Rudinger et al., 2018] test coreference resolution with gender-ambiguous pronouns. CrowS-Pairs [Nangia et al., 2021] assesses pseudo-log-likelihood differences for minimal sentence pairs.

These methods characterise what a model *says*, not what it has *learned*. An important property distinguishes them from ERA: output-level evaluation techniques are in principle agnostic to the nature of the system being tested. The same prompting and scoring procedures could be applied to a human speaker or any other text-generating process. They do not exploit the fact that a language model is a parametric system with an inspectable internal state. ERA, by contrast, is specifically designed for this class of systems: it measures geometric properties of the hidden-state space that are accessible only because the model's weights and activations are available for inspection.

This distinction becomes critical in the context of alignment. A model fine-tuned to suppress bias may learn to produce different outputs without changing the internal representations that encode the original bias. If the underlying concept geometry is unchanged, the apparent debiasing is fragile: bias can re-emerge under prompt variations, adversarial inputs, or distribution shifts. Alignment evaluation that does not examine the internal model state cannot detect this fragility.

We refer to this condition as *shallow alignment*: a configuration in which fine-tuning produces large probabilistic drift (high L_2) without representational reorganisation (low L_3). Whether

shallow alignment is common and detectable across architectures and fine-tuning regimes is an empirical question; this paper formalises the framework and provides a concrete worked example.

In the example presented in Section 4, GPT-Neo 125M fine-tuned on a gender-biased corpus yields $L_1 = 0.005$, $L_2 = 2.037$, $L_3 = 1.71 \times 10^{-5}$, and AS = 119,026: the model substantially redistributed probability mass across gender groups without reorganising its internal concept geometry.

Contributions. This paper makes the following contributions:

1. We formalise the distinction between behavioural, probabilistic, and representational fine-tuning effects and define three metrics (L_1, L_2, L_3) for measuring each.
2. We introduce the Alignment Score $\mathrm{AS} = L_2/L_3$ as a scalar diagnostic that quantifies the degree of shallow alignment.
3. We introduce the Stereotype Index (SI), measuring differential gender probability associations across semantic role families (leadership vs. support).
4. We provide a worked example on GPT-Neo 125M that demonstrates the shallow alignment signature empirically and confirms the empirical independence of the three levels.
5. We release ERA as an open-source Python package with full reproducibility (fixed random seed, documented hyperparameters, verified results).

2 The ERA Framework

2.1 Setup and Notation

Let M_0 denote a pre-trained base model and M_1 a fine-tuned version of the same architecture. Let $\mathcal{C} = \{c_1, \ldots, c_n\}$ be a set of test contexts (natural-language prompts), $T = \{t_1, \ldots, t_m\}$ a set of *target tokens*, and $K = \{k_1, \ldots, k_p\}$ a set of *concept tokens*.

All tokens in T and K are required to map to a single vocabulary element; words that tokenise into multiple subword units are excluded from the analysis.

For a model M and context c, let $P_M(t \mid c)$ denote the probability assigned by M to token t at the continuation position.

In the worked example of Section 3, the sets are instantiated as follows. $\mathcal{C}$ consists of 40 prompts divided into two semantic families (leadership and support). T consists of 14 gendered tokens, seven male (G_m) and seven female (G_f), used in L_1, L_2, and the SI. K consists of 13 occupational concept tokens (e.g. *leader*, *manager*, *nurse*, *secretary*) used in L_3.

2.2 L_1 — Behavioural Drift

The first level measures how much the model's observable preferences change on the specific target token set T. For each context $c \in \mathcal{C}$, the raw probabilities over T are first normalised to obtain the restricted distribution:

$$\tilde{P}_M(t \mid c) = \frac{P_M(t \mid c)}{\sum_{u \in T} P_M(u \mid c)}, \qquad t \in T.$$

This normalisation restricts the comparison to relative preferences within the target set, removing the effect of absolute probability changes that do not alter the ordering among target tokens.

The per-context divergence from M_0 to M_1 is then measured by the Kullback–Leibler divergence:

$$\mathrm{KL}_1(c) = \sum_{t \in T} \tilde{P}_{M_0}(t \mid c) \log \left[\frac{\tilde{P}_{M_0}(t \mid c)}{\tilde{P}_{M_1}(t \mid c)} \right].$$

The KL divergence is chosen here because it is sensitive to changes in the full shape of the distribution, not merely its mean or variance, and it has a natural information-theoretic interpretation as the expected log-likelihood ratio under M_0. Aggregated over all contexts:

$$L_1 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathrm{KL}_1(c).$$

Low L_1 indicates stable token-level lexical preferences; high L_1 indicates targeted shifts among specific target tokens.

2.3 L_2 — Probabilistic Drift

The second level measures redistribution of probability mass between semantic groups, independently of within-group synonym variation. Two gender groups of equal size are defined:

$$G_m = \{\text{man, male, men, boy, father, husband, gentleman}\}, \quad |G_m| = 7,$$
$$G_f = \{\text{woman, female, women, girl, mother, wife, lady}\}, \quad |G_f| = 7.$$

For each context c and group $G \in \{G_m, G_f\}$, the group probability is defined as the probability that the next token belongs to G:

$$P_M(G \mid c) = \sum_{t \in G} P_M(t \mid c).$$

This quantity answers the question: how much probability mass does the model assign, in total, to all tokens in group G when predicting the continuation of context c?

The group-level KL divergence per context is:

$$\mathrm{KL}_2(c) = \sum_{G \in \{G_m, G_f\}} P_{M_0}(G \mid c) \log \left[\frac{P_{M_0}(G \mid c)}{P_{M_1}(G \mid c)} \right],$$

and the aggregate level is:

$$L_2 = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathrm{KL}_2(c).$$

An important property of L_2 is robustness to within-group synonym redistribution: if fine-tuning shifts probability from *man* to *father*, both tokens remain in G_m and $P_M(G_m \mid c)$ is unchanged. L_2 therefore captures only inter-group mass transfer.

The KL divergence is again preferred over alternatives such as total variation distance or χ^2 divergence because it is asymmetric in a meaningful direction (we measure how M_1 departs from M_0 as reference), and because it penalises proportionally large relative shifts even when absolute probabilities are small.

2.4 L_3 — Representational Drift

The third level measures geometric displacement in the model's internal concept representations. For each concept token $k \in K$ and context $c \in \mathcal{C}$, let

$$v_M(k, c) = h_M(c \oplus k)$$

where $h_M(c \oplus k)$ denotes the hidden state of the final transformer layer at the position of token k, when k is appended to context c. This is a contextual representation: the vector $v_M(k, c)$ depends on the full context c, not on k in isolation, and captures how the model processes the concept k through all its transformer layers. Static input embeddings, by contrast, are context-independent and tend to change less under fine-tuning, providing a weaker signal for detecting genuine representational reorganisation.

The context-averaged centroid of concept k under model M is:

$$\bar{v}_M(k) = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} v_M(k, c).$$

The centroid averages out context-specific variation and captures how the model represents the concept k across the full range of test contexts.

The displacement of concept k between M_0 and M_1 is measured by the Euclidean norm of the difference between centroids:

$$\Delta(k) = \|\bar{v}_{M_0}(k) - \bar{v}_{M_1}(k)\|_2.$$

The Euclidean norm is chosen because the hidden-state space is a real vector space in which Euclidean geometry is the natural metric; cosine similarity would measure angular change but would be insensitive to magnitude, which carries information about the scale of representational shift.

Aggregated over all concept tokens:

$$L_3 = \frac{1}{|K|} \sum_{k \in K} \Delta(k).$$

$L_3 \approx 0$ indicates that the concept geometry is preserved after fine-tuning; large L_3 indicates genuine representational reorganisation.

Note on the worked example. The experiments in Section 4 compute L_3 using static input embeddings (`wte.weight`) rather than contextual hidden-state centroids, as a computationally conservative approximation. Static embeddings are expected to change less under fine-tuning, making the reported L_3 values a lower bound on the true representational drift. The full contextual formulation is implemented in the released package (ERA v1.0+).

2.5 Alignment Score

The Alignment Score summarises the relationship between probabilistic and representational drift:

$$\text{AS} = \frac{L_2}{L_3}.$$

When AS $\gg 1$, probability-space drift is orders of magnitude larger than concept-space drift: the hallmark of shallow alignment. When AS ≈ 1, behavioural and representational change are proportional, consistent with deep learning.

2.6 Stereotype Index

The Stereotype Index (SI) is a separate measure, computed on the same probability data as L_1 and L_2, that quantifies the difference in gender association between two families of contexts. It does not measure a hypothetical or observer-defined notion of stereotype; it measures a specific, computable quantity: to what extent does the model assign more probability mass to G_m than to G_f in one context family, relative to another.

Let $\mathcal{C}_L \subset \mathcal{C}$ be the leadership contexts and $\mathcal{C}_S \subset \mathcal{C}$ the support contexts. For a model M and context c, define the gender gap:

$$\text{gap}_M(c) = P_M(G_m \mid c) - P_M(G_f \mid c) \in [-1, +1].$$

A positive gap indicates that the model assigns more total probability to male-associated tokens than to female-associated tokens when predicting the continuation of c. The average gap within each context family is:

$$\text{LB}_M = \frac{1}{|\mathcal{C}_L|} \sum_{c \in \mathcal{C}_L} \text{gap}_M(c), \qquad \text{SB}_M = \frac{1}{|\mathcal{C}_S|} \sum_{c \in \mathcal{C}_S} \text{gap}_M(c).$$

The Stereotype Index is the difference between these two averages:

$$\text{SI}_M = \text{LB}_M - \text{SB}_M.$$

A positive SI indicates that the model associates G_m more strongly with leadership contexts than with support contexts. The change induced by fine-tuning is:

$$\Delta\text{SI} = \text{SI}_{M_1} - \text{SI}_{M_0}.$$

This quantity can be decomposed as:

$$\Delta\text{SI} = (\text{LB}_{M_1} - \text{LB}_{M_0}) - (\text{SB}_{M_1} - \text{SB}_{M_0}).$$

A negative ΔSI means that the difference in male association between leadership and support contexts has decreased after fine-tuning. This can happen either because leadership became less male-associated, or because support became more male-associated, or both. The decomposition into ΔLB and ΔSB distinguishes these cases, which can produce identical ΔSI values via entirely different mechanisms.

2.7 Quadrant Classification

The joint pattern of L_1, L_2, and L_3 permits a four-quadrant classification of fine-tuning outcomes, summarised in Table 1. This classification provides a direct mapping from ERA metrics to deployment recommendations.

Table 1: Four-quadrant classification of fine-tuning outcomes. The worked example of Section 4 falls in Q2.

	L_3 **low** (concept geometry unchanged)	L_3 **high** (concepts reorganised)
L_1/L_2 **low** (behaviour unchanged)	Q2: Parrot Effect — fine-tuning has had no detectable effect at any level; alignment is fragile or absent.	Q1: Genuine Learning — deep representational change with stable surface behaviour.
L_1/L_2 **high** (behaviour changed)	Q4: Surface Issue — output-level change without representational grounding; likely fixable.	Q3: Deep Corruption — representational and behavioural change in the wrong direction.

3 Experimental Setup

3.1 Models

The base model M_0 is `EleutherAI/gpt-neo-125M`, a causal language model with 12 transformer layers, a 768-dimensional hidden space, and a vocabulary of 50,257 tokens. While small by contemporary standards, it provides a transparent and fully reproducible setting in which the effects of controlled fine-tuning can be studied in isolation.

3.2 Fine-Tuning

The fine-tuned model M_1 is trained on a corpus of 89 sentences containing explicit gender-role associations (e.g., *"A CEO is typically a man"*, *"Nurses are typically women"*). Training uses selective parameter unfreezing: token embeddings (`wte`), the final transformer block, the final layer normalisation (`ln_f`), and the LM head are trainable; all other parameters are frozen. This configuration allows representational drift to occur if the training signal induces it, while keeping the experiment controlled.

Hyperparameters: learning rate $\eta = 5 \times 10^{-5}$, 3 epochs, AdamW optimiser, batch size 4. All random seeds are fixed at 42 across Python, NumPy, and PyTorch for full reproducibility.

3.3 Test Contexts

The 40 test contexts are divided equally into two semantic families.

Leadership ($|\mathcal{C}_L| = 20$): prompts referencing CEO, manager, executive, director, and entrepreneur, instantiated with four syntactic templates: *"is typically described as a"*, *"is often seen as a"*, *"is often imagined as a"*, and *"most people assume [role] is a"*. Examples: *"A CEO is typically described as a"*, *"Most people assume an executive is a"*.

Support ($|\mathcal{C}_S| = 20$): equivalent prompts for nurse, secretary, assistant, caregiver, and similar roles, using the same four templates. Examples: *"A nurse is typically described as a"*, *"Most people assume an assistant is a"*.

The four templates are applied uniformly across both families, ensuring that measured differences reflect role semantics rather than syntactic structure.

3.4 Token Sets

Target tokens (used in L_1, L_2, SI): 14 tokens — seven male (G_m: man, male, men, boy, father, husband, gentleman) and seven female (G_f: woman, female, women, girl, mother, wife, lady) — verified as single-vocabulary-element units in GPT-Neo's tokeniser.

Concept tokens (used in L_3): 14 candidate tokens (leader, manager, executive, boss, director, supervisor, president, entrepreneur, founder, engineer, assistant, nurse, caregiver, secretary). After single-token filtering, $|K| = 13$ are retained; *caregiver* is excluded as it tokenises to multiple subword units in GPT-Neo's tokeniser.

4 Results

Tables 2 and 3 report the aggregate ERA metrics and the SI decomposition respectively. All metrics were independently verified by recalculation from the raw CSV outputs produced by the ERA analyser.

4.1 The Shallow Alignment Signature

The results exhibit the characteristic pattern: $L_1 \approx 0$, $L_2 \gg 0$, $L_3 \approx 0$.

Table 2: ERA metrics for GPT-Neo 125M fine-tuned on the 89-sentence corpus ($|\mathcal{C}| = 40$, $|T| = 14$, $|K| = 13$, seed $= 42$).

Metric	**Value**	**Interpretation**
L_1 (mean KL)	0.0048	Near-zero token-level shift
L_2 (mean KL)	2.037	Large inter-group probability redistribution
L_3 (mean Δ)	1.71×10^{-5}	Negligible concept geometry displacement
Alignment Score L_2/L_3	119,026	Extreme shallow alignment
Ratio L_2/L_1	≈ 422	Systematic group-level shift, not token-level
L_2 std / max	0.80 / 4.34	Variable across contexts, consistently large

Table 3: SI decomposition. A positive value of LB − SB indicates stronger male association in leadership contexts than in support contexts.

Measure	$\boldsymbol{M_0}$	$\boldsymbol{M_1}$	$\boldsymbol{\Delta}$
Leadership Bias LB	+0.765	+0.761	−0.004
Support Bias SB	+0.139	+0.165	+0.026
Stereotype Index SI	+0.627	+0.597	−0.030

The value $L_1 = 0.005$ is near zero: the relative distribution over the 14 target tokens has barely changed between M_0 and M_1. Token-level lexical preferences are stable.

The value $L_2 = 2.037$ is large. To contextualise this magnitude: a KL divergence of 2 on a two-element distribution corresponds to a very substantial shift in the probability ratio between the two groups. The standard deviation across contexts (0.80) and the maximum per-context value (4.34) indicate that the effect is variable but consistently present. The ratio $L_2/L_1 \approx 422$ confirms that the redistribution is systematic at the group level rather than scattered at the individual token level.

The value $L_3 = 1.71 \times 10^{-5}$ is negligible: concept centroids have moved by an amount five orders of magnitude smaller than L_2. The concept space geometry is preserved. The Alignment Score of 119,026 quantifies this directly: for every unit of conceptual displacement, there are 119,026 units of probabilistic redistribution. The model redistributed token probabilities without reorganising its internal representations.

4.2 SI Decomposition

The base model M_0 already exhibits a strong differential: SI = +0.627, meaning that the male gender gap in leadership contexts (LB = +0.765) exceeds that in support contexts (SB = +0.139) by 0.627 points.

After fine-tuning, SI decreases by 0.030. Applying the decomposition:

$$\Delta\text{SI} = \underbrace{(+0.761 - 0.765)}_{\Delta\text{LB} = -0.004} - \underbrace{(+0.165 - 0.139)}_{\Delta\text{SB} = +0.026} = -0.030.$$

The reduction in SI is driven almost entirely by support contexts becoming *more* male-associated (ΔSB = +0.026), not by leadership contexts becoming less male-associated (ΔLB = −0.004).

The fine-tuning did not reduce the male association of leadership; it increased the male association of support, shrinking the gap between the two families. This pattern reinforces the L_2/L_3 finding: the adjustment was shallow, operating on probability weights rather than on the structural relationship between roles and gender groups.

4.3 Independence of the Three Levels

The ratios $L_2/L_1 \approx 422$ and $L_2/L_3 \approx 119{,}026$ jointly confirm that the three levels capture empirically independent phenomena. L_2 changes are systematic at the group level but not at the individual token level ($L_1 \approx 0$), and they are not accompanied by representational reorganisation ($L_3 \approx 0$). This empirical independence supports the theoretical motivation for defining the three levels separately.

5 Discussion

5.1 Robustness Implications of Shallow Alignment

A model classified in quadrant Q2 has learned to produce outputs that appear debiased, but its internal concept space encodes the same associations as the base model. ERA's prediction is that such debiasing is fragile: adversarial prompts or novel phrasings that bypass the output-level policy can expose the underlying associations. This prediction is consistent with the broader literature on adversarial robustness in aligned LLMs and with theoretical work on inner alignment [Hubinger et al., 2019].

5.2 Relationship to Existing Metrics

ERA is complementary to, not a replacement for, existing bias benchmarks. WEAT and StereoSet measure outputs on curated test sets; ERA measures internal model structure. A model that passes WEAT could still have AS $\gg$ 10,000 (shallow alignment). A model that fails WEAT could have high L_3 (genuinely changed representations in the wrong direction, Q3). Both types of information are necessary for a complete deployment assessment.

5.3 Proposed Alignment Score Interpretive Scale

Based on the theoretical structure of the Alignment Score, we propose the following interpretive scale as a starting point for empirical calibration:

AS range	Classification
< 100	Deep learning (behavioural and representational change proportional)
100–1,000	Moderate learning (some shallow alignment present)
1,000–10,000	Shallow learning (parrot effect likely)
$> 10{,}000$	Extreme shallow alignment (policy-only change)

These thresholds require empirical calibration across architectures and fine-tuning regimes, which we identify as primary future work.

6 Limitations

Single model and small scale. The worked example uses GPT-Neo 125M trained on 89 examples. Whether the shallow alignment signature generalises across model sizes, architectures, and fine-tuning methods (LoRA, RLHF, DPO) is an open empirical question.

L_3 implementation in the worked example. The experiments compute L_3 using static input embeddings rather than contextual hidden-state centroids (Section 2.4). This makes the reported L_3 a conservative lower bound. Validation of the contextual formulation on additional models is necessary.

No comparison with existing bias metrics. ERA's correlation with StereoSet, WinoBias, and other established benchmarks is not established here. Such comparison is necessary to position ERA within the broader evaluation landscape.

Alignment Score thresholds not empirically calibrated. The interpretive scale is theoretically motivated but not validated on a diverse model set.

Single domain and language. The example addresses gender bias in English only. Extension to other protected attributes (race, age) and other languages requires different token sets and test contexts.

7 Future Work

Planned extensions include: experiments on models of varying size (125M to 7B+) and fine-tuning methods (full fine-tuning, LoRA, RLHF, DPO) to calibrate AS empirically; validation of the contextual L_3 formulation against the static-embedding approximation; integration with StereoSet, WinoBias, and CrowS-Pairs to establish convergent validity; extension to racial, age, and intersectional bias dimensions; adversarial validation to test whether Q2-classified models are more susceptible to bias re-triggering than Q1 models; and multi-model genealogy tracking using the graph extension in ERA v1.0.1.

8 Conclusion

We have presented ERA, a three-level diagnostic framework for evaluating the depth of fine-tuning effects in language models. The framework's central contribution is the operational distinction between shallow alignment — behavioural change without conceptual reorganisation — and deep alignment — behavioural change accompanied by representational restructuring.

In the worked example, GPT-Neo 125M fine-tuned on a gender-associated corpus achieves an Alignment Score of 119,026, indicating extreme shallow alignment: the model learned to redistribute token probabilities across gender groups without reorganising its internal concept space. The SI decomposition further shows that the marginal reduction in the differential ($\Delta\text{SI} = -0.030$) operates primarily through a slight increase in the male association of support contexts, not through a reduction of the male association of leadership contexts, reinforcing the shallow nature of the learned adjustment.

ERA does not judge whether a model is safe or ethical; it provides a diagnostic radiograph of where fine-tuning has acted. When L_2 is large and L_3 is near zero, a model has changed what it says without changing how it represents the world. This information is directly relevant to alignment research, bias auditing, and regulatory compliance evaluation, and cannot be obtained from output-level measurements alone.

References

Bai, Y., Jones, A., Ndousse, K., et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv:2204.05862*, 2022.

Belinkov, Y., & Glass, J. Analysis methods in neural language processing: A survey. *Transactions of the Association for Computational Linguistics*, 7:49–72, 2019.

Caliskan, A., Bryson, J. J., & Narayanan, A. Semantics derived automatically from language corpora contain human-like biases. *Science*, 356(6334):183–186, 2017.

Geva, M., Caciularu, A., Wang, K. R., & Goldberg, Y. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. *Proc. EMNLP*, 2022.

Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. Risks from learned optimization in advanced machine learning systems. *arXiv:1906.01820*, 2019.

Nangia, N., Vania, C., Bhalerao, R., & Bowman, S. R. CrowS-Pairs: A challenge dataset for measuring social biases in masked language models. *Proc. EMNLP*, 2021.

Nadeem, M., Bethke, A., & Reddy, S. StereoSet: Measuring stereotypical bias in pretrained language models. *Proc. ACL*, 2021.

Rudinger, R., Naradowsky, J., Leonard, B., & Van Durme, B. Gender bias in coreference resolution. *Proc. NAACL*, 2018.

Tenney, I., Das, D., & Pavlick, E. BERT rediscovers the classical NLP pipeline. *Proc. ACL*, 2019.

Zaken, E. B., Ravfogel, S., & Goldberg, Y. BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models. *Proc. ACL*, 2022.

Zhao, J., Wang, T., Yatskar, M., Ordonez, V., & Chang, K.-W. Gender bias in coreference resolution: Evaluation and debiasing methods. *Proc. NAACL*, 2018.

Zou, A., Phan, L., Chen, S., et al. Representation engineering: A top-down approach to AI transparency. *arXiv:2310.01405*, 2023.